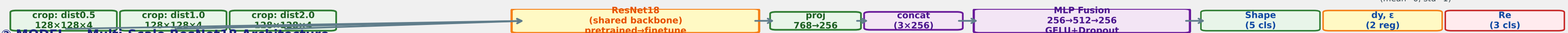
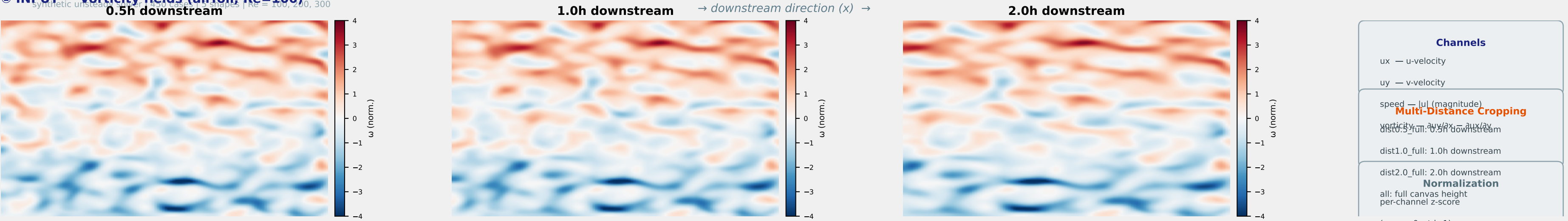
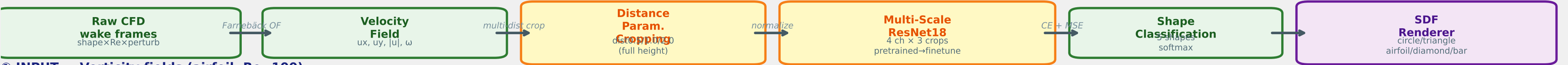


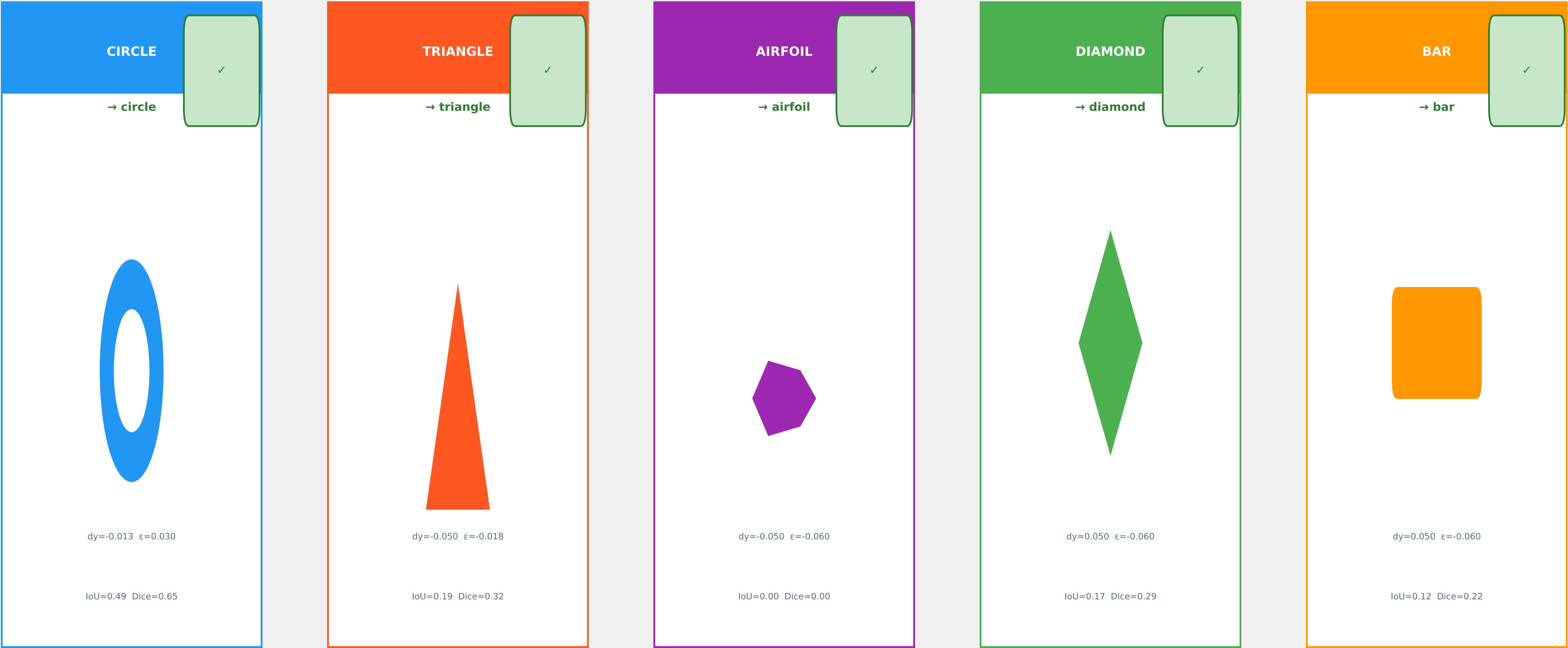
Wake-Field Shape Classification Pipeline

Input: downstream velocity fields at configurable distances → Model: multi-scale ResNet18 → Output: obstacle shape + position parameters



③ **OUTPUT — Shape Predictions (distance-param. multi-scale model)**

Training: 8 epochs · batch=16 · lr=8e-4 · weight_decay=1e-4 · multi-task loss: CE(shape) + 0.5·MSE(params) + 0.2·CE(Re)



Accuracy=91.1% Macro-F1=90.9% | dy MAE=0.040 ε MAE=0.052 | Inverse IoU=0.38 Dice=0.49 | 450 cases · 5 shapes · Re∈{100,200,300}